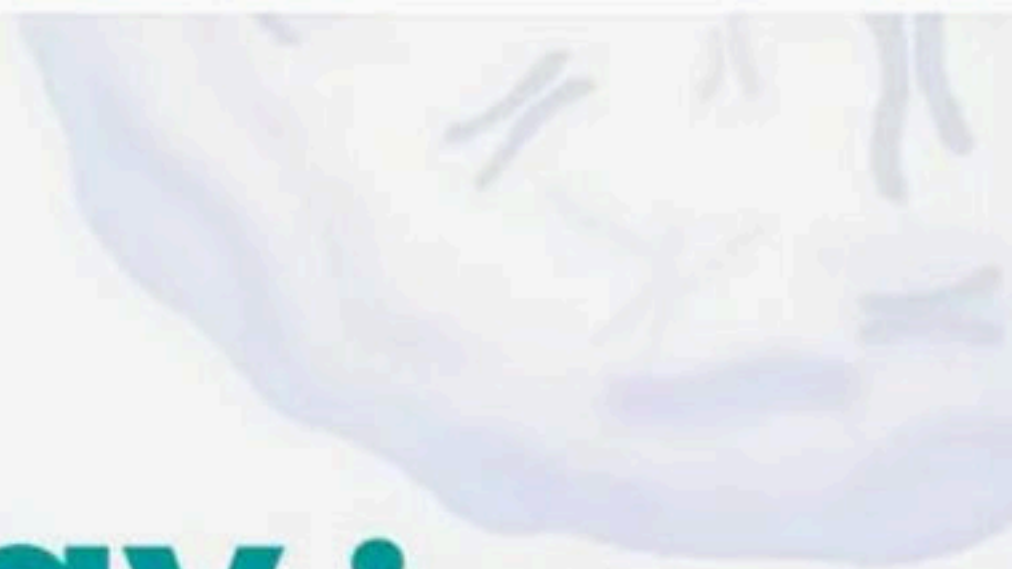


Biotechnology : Principle & Processes - 100 MCQs

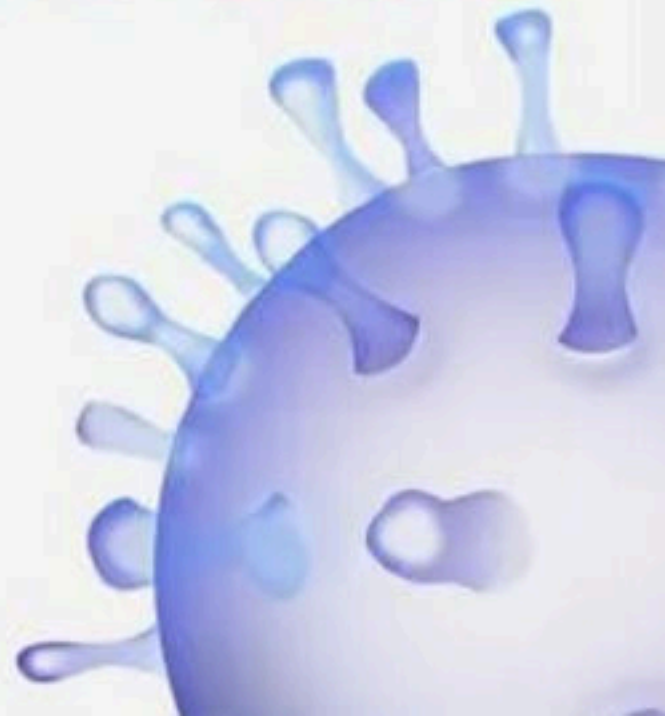
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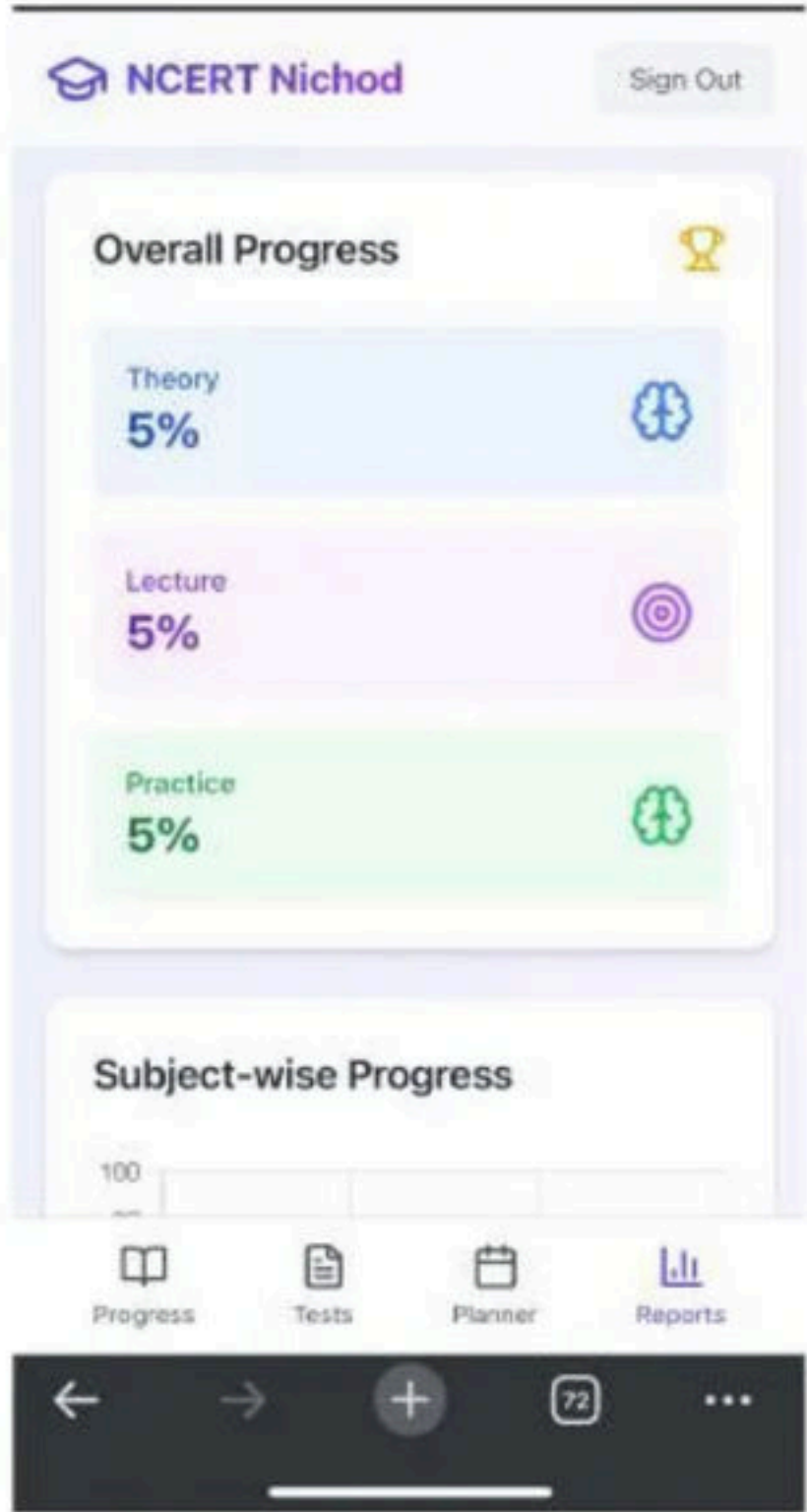
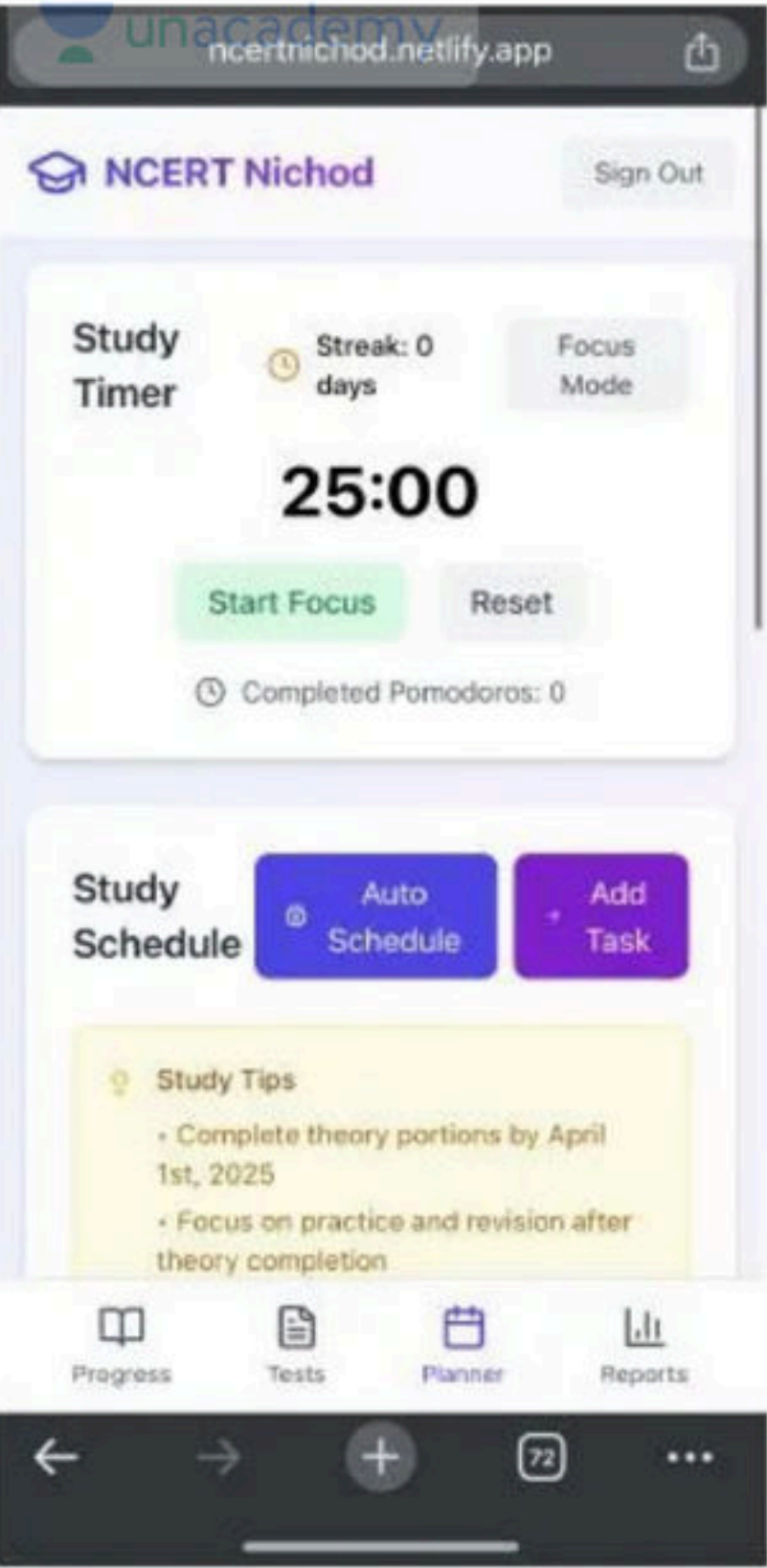
Tests



Planner



Reports



1. Consider the given statements w.r.t. gel electrophoresis.

I. DNA can be separated by forcing them to move towards anode under an electric field.

II. DNA fragments separate according to their size through sieving effect provided by agarose gel.

III. The used matrix in this technique is agarose which is an artificial polymer.

Which of the above statements is/ are correct?

(1) II only

(2) I and II

(3) I, II, and III

(4) II and III



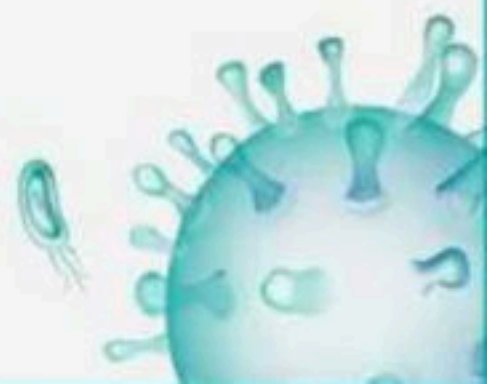
2. How many restriction enzymes are isolated till now?

- (1) 90
- (2) 9000
- (3) 900
- (4) 230



3. The enzymes most commonly used in genetic engineering are

- (1) Restriction endonuclease and RNA polymerase
- (2) Endonuclease and ligase
- (3) Restriction endonuclease and ligase
- (4) Ligase and phosphatases



4. Which of the following statements is incorrect?

- (1) A bio-reactor provides optimal growth conditions for achieving the desired product
- (2) Most commonly used bio-reactors are of stirring type
- (3) Bio-reactors are used to produce small scale bacterial cultures
- (4) Bio-reactors have an agitator system, an oxygen delivery system and foam control system



5. Assertion : In recombinant DNA technology, human genes are always transferred into bacteria (Prokaryotes) or yeast (eukaryotes)

Reason : Both bacteria and yeast multiply very fast to form huge populations which express the desired gene

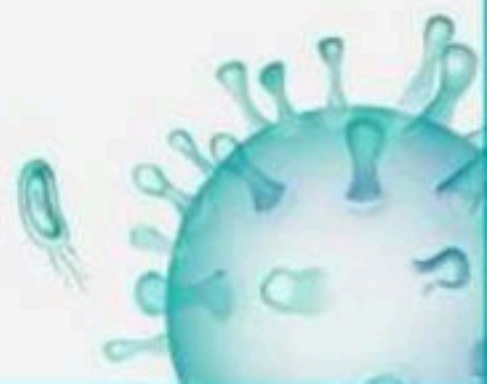
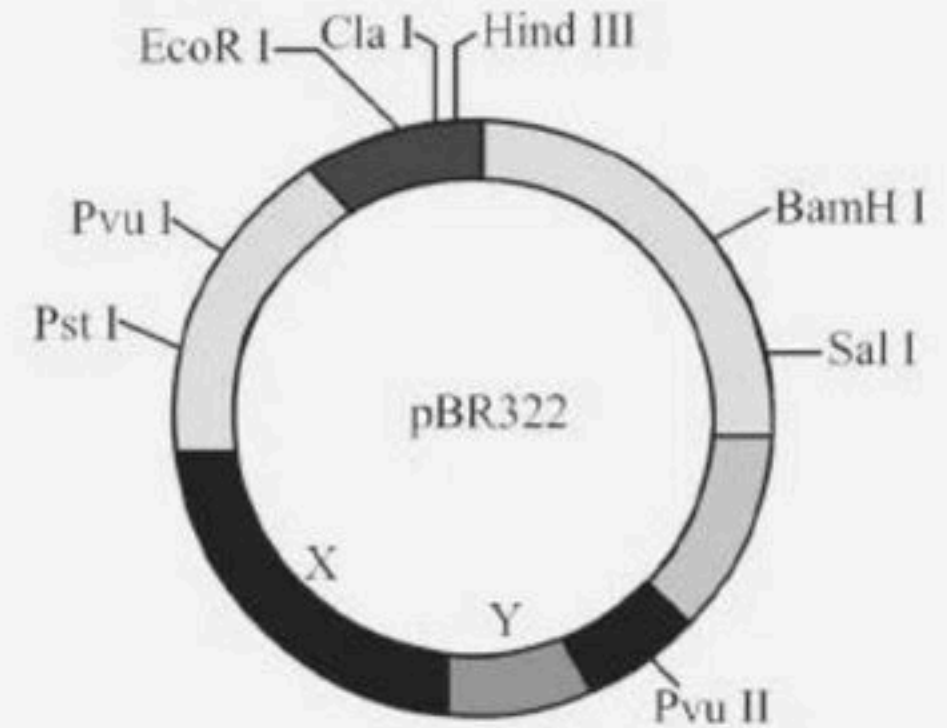
- (1) Both Assertion and Reason are correct and Reason is the correct explanation of the Assertion.
- (2) Both Assertion and Reason are correct but Reason is not the correct explanation of the Assertion.
- (3) Assertion is correct but Reason is incorrect.
- (4) Both Assertion and Reason are incorrect



6. The following diagram showing restriction sites in E. coli cloning vector pBR322.

Find the role of ' X ' and ' Y ' genes :

- (1) The gene ' X ' is responsible for resistance to antibiotics and ' Y ' for protein involved in the replication of Plasmid.
- (2) The gene ' X ' is responsible for controlling the copy number of the linked DNA and ' Y ' for protein involved in the replication of Plasmid.
- (3) The gene ' X ' is for protein involved in replication of Plasmid and 'Y' for resistance to antibiotics.
- (4) Gene ' X ' is responsible for recognitions sites and ' Y ' is responsible for antibiotic resistance.



7. The core techniques that gave birth to modern biotechnology are

I. Genetic engineering

II. Bioprocess engineering

III. HGP

IV. Bioinformatics

Choose the correct option

(1) I and II

(2) I and III

(3) II and IV

(4) II and III



8. In a genetic engineering experiment, restriction enzymes can be used for

- (1) Prokaryotic DNA only
- (2) Viral DNA only
- (3) Any DNA fragment
- (4) Eukaryotic DNA only



9. The 'Ori' region enables plasmid

- (1) Undergo transformation
- (2) Replication in host cell
- (3) Responsible for controlling the copy number of the linked DNA
- (4) Both 2 and 3



10. During the purification process for recombinant DNA technology, addition of chilled ethanol precipitates out

- (1) RNA
- (2) DNA
- (3) Histones
- (4) Polysaccharides



11. PCR is used for

- (1) DNA ligation
- (2) DNA digestion
- (3) DNA amplification
- (4) DNA isolation



12. Statement A : pB322 is a plasmid vector.

Statement B : A plasmid is an extrachromosomal genetic elements of DNA that is capable of replicating independently of the host chromosome.

- (1) Statement A is correct but Statement B is incorrect.
- (2) Statement A is incorrect but Statement B is correct.
- (3) Both statements are correct.
- (4) Both Statements are incorrect.



13. In restriction endonuclease called EcoRI, what does 'R' part stands for?

- (1) Name of the strain
- (2) Name of the genus
- (3) Name of the species
- (4) Name of the class



14. Which statement is correct?

- I. A large scale production of desired biotechnological product is done by using Bioreactors**
- II. Downstream processing is one of the steps of RDNA technology.**
- III. Bioreactors are vessels of large volumes in which raw materials are biologically converted into specific products.**
- IV. The presence of chromogenic substrate gives blue colour colonies, if the plasmid in the bacteria does not have an insert.**

- (1) I and II
- (2) I, III and IV
- (3) I and III
- (4) None of these



15. Select the correct answer

A. Restriction enzyme digestions are performed by incubating purified DNA molecules with the restriction enzyme, at the optimal conditions for that specific enzyme.

B. Agarose gel electrophoresis is employed to check the progression of a restriction enzyme digestion.

(1) Only A is correct

(2) Only B is correct

(3) Both are correct

(4) None are correct



16. Which one is a true statement regarding DNA polymerase used in PCR?

- (1) It serves as a selectable marker
- (2) It is isolated from a virus
- (3) It remains active at high temperature
- (4) It is used to ligate introduced DNA in recipient cell



17. EFB stands for

- (1) European Federation of Biotechnology
- (2) European Foundation of Biotechnology
- (3) European Force of Biotechnology
- (4) European Federation of Biological sciences



18. In gel electrophoresis, DNA fragments separate according to their size through____ provided by the agarose gel.

- (1) Elution
- (2) Sieve effect
- (3) Spooling
- (4) Transformation



19. Plasmids are suitable vectors for genetic cloning as

- (1) They are indispensable
- (2) They are self replicating units
- (3) They are essential for bacterial reproduction
- (4) None of the above



20. Match the columns I and II regarding competent host and the technique used

Column I

A Biolistics

B Micro-injection

C Heat shock

Column II

P Animal host

Q Bacterial host

R Plant host

(1) A – R , B – P , C – Q

(2) A – Q , B – R , C – P

(3) A – P , B – R , C – Q,

(4) A – Q , B – P , C – R



21. Colonies of recombinant bacteria appear white in contrast to blue colonies of non recombinant bacteria because

- (1) Insertional inactivation of alpha galactosidase in nonrecombinant bacteria.
- (2) Non recombinant bacteria containing beta galactosidase
- (3) Insertional inactivation of alpha galactosidase in recombinant bacteria.
- (4) Inactivation of glycosidase enzyme in recombinant bacteria.



22. Consider the following statements

I. Source of the restriction enzyme Hind III is Helicobacter influenzae

II. DNA is a hydrophilic molecule

III. Separated DNA fragments are visualized after staining the DNA with ethanol and using UV light

IV. Presence of more than one same recognition sites within vector will generate several fragments of vector, which will complicate gene cloning

Which of the above statement are correct?

(1) I, II & III

(2) II, III & IV

(3) II & IV

(4) I, II, & IV



23. When an foreign DNA is ligated in tetracycline resistant gene, the recombinant

- (1) Become tetracyclic resistant
- (2) Will loose tetracyclic resistant
- (3) Become ampicillin resistant
- (4) None of these



24. Assertion : Restriction enzymes Hind II and Hpa are produced from two different genera of bacteria.

Reason : Hind II is produced from Haemophilus while Hpa is produced from Haematococcus.

- (1) Both Assertion and Reason are correct and Reason is the correct explanation of the Assertion.
- (2) Both Assertion and Reason are correct but Reason is not the correct explanation of the Assertion.
- (3) Assertion is correct but Reason is incorrect.
- (4) Both Assertion and Reason are incorrect



25. Given below are two statements :

Statement A : Restriction endonucleases recognize specific sequence to cut DNA known as palindromic nucleotide sequence.

Statement B : Restriction endonucleases cut the DNA strand a little away from the centre of the palindromic site.

In the light of the above statements, choose the most appropriate answer from the options given below :

- (1) Both Statement A and Statement B are incorrect
- (2) Statement A is correct but Statement B is incorrect
- (3) Statement A is incorrect but Statement B is correct
- (4) Both Statement A and Statement B are correct



26. Restriction enzyme cuts the DNA strand a little away from the centre of palindrome site between

- (1) Same two bases on same strand
- (2) Opposite bases on same strand
- (3) Same two bases on opposite strand
- (4) Opposite bases on opposite strand



27. Identify the palindromic sequence in the following base sequences.



28. Elution means:

- (1) Separation of DNA fragments on agarose gel
- (2) Cutting and extraction of DNA bands from the agarose gel
- (3) Making the DNA bands visible under UV radiation
- (4) Isolating alien DNA from the choice organism



29. In insertional inactivation, the recombinant DNA is inserted within the coding sequence of

- (1) Ampicillin resistant gene
- (2) Tetracycline resistant gene
- (3) Kanamycin resistant gene
- (4) β -galactosidase



30. The plasmids and bacteriophages have the ability to replicate within ____ A ____ cells independent of the control of ____ B ____ DNA.

- (1) A - bacterial, B - plasmid
- (2) A - bacterial, B - chromosomal
- (3) A - viral, B - chromosomal
- (4) A - viral, B - plasmid



31. Stirred tank bioreactor have been designed for

- (1) Addition of preservatives to the product
- (2) Purification of the product
- (3) Availability of oxygen throughout the process
- (4) Ensuring anaerobic conditions in the cultural vessel



32. Restriction endonucleases are

- (1) Present in mammalian cells for degradation of DNA when a cell dies.
- (2) Used in genetic engineering for ligating two molecules
- (3) Synthesized by bacteria as a part of their defense mechanism
- (4) Used in invitro DNA synthesis



33. Selection of recombinants due to inactivation of antibiotics is a cumbersome procedure because

- (1) It requires plating of one plate having certain antibiotics
- (2) It requires plating of two plates having same antibiotics
- (3) It requires simultaneous plating on two plates having different antibiotics
- (4) None of these



34. Genetic engineering is possible because

- (1) DNA can be visualised by electron microscope
- (2) DNA ligase can cut DNA at specific sites
- (3) Restriction endonucleases can be purified from bacteria can be used in vitro
- (4) We can cut DNA at all sites by endonucleases



35. Restriction endonucleases are isolated from some bacteria. Their role in bacteria is:

- (1) Defense against virus
- (2) Synthesis of proteins
- (3) Act as genetic material
- (4) Help in reproduction



36. Assertion : PCR technique is helpful in detecting bacterial and viral diseases even when symptoms of disease are not yet visible

Reason : Very low concentrations of bacteria and viruses in human body can be detected by amplification of the nucleic acids using PCR technique.

- (1) Both Assertion and Reason are correct and Reason is the correct explanation of the Assertion.
- (2) Both Assertion and Reason are correct but Reason is not the correct explanation of the Assertion.
- (3) Assertion is correct but Reason is incorrect.
- (4) Both Assertion and Reason are incorrect



37. The recognition site for Bam in is present in

- (1) Rop site
- (2) Tetracycline resistant site
- (3) Ori site
- (4) Ampicillin resistant site



38. Select the correct answer

I. Strict quality control testing for each product is also required.

II. The downstream processing and quality control testing vary from product to product

- (1) Only I is correct
- (2) Only II is correct
- (3) Both are correct
- (4) None are correct



39. Which of the following are the part of biotechnology?

i. Synthesis of gene

ii. Developing a DNA vaccine

iii. Correcting a defective gene

iv. In vitro fertilization

(1) , ii, iii

(2) i and ii

(3) iii and iv

(4) All of them



40. Statement A : Biotechnology deals with techniques that use living organism to produce products useful for humans.

Statement B : Construction of recombinant DNA involves cleaving DNA segments with endonuclease and rejoining with ligase.

- (1) Statement A is correct but Statement B is incorrect.
- (2) Statement A is incorrect but Statement B is correct.
- (3) Both statements are correct.
- (4) Both Statements are incorrect.



41. Assertion : All endonucleases cut DNA at specific sites

Reason : Endonucleases are found in viruses

- (1) Both Assertion and Reason are correct and Reason is the correct explanation of the Assertion.
- (2) Both Assertion and Reason are correct but Reason is not the correct explanation of the Assertion.
- (3) Assertion is correct but Reason is incorrect.
- (4) Both Assertion and Reason are incorrect



42. Insertional inactivation is related to

- (1) Gene gun
- (2) Microinjection
- (3) Selection of recombinants
- (4) Gel electrophoresis



43. The DNA fragments separated on an agarose gel can be visualised after staining with :

- (1) Acetocarmine
- (2) Aniline blue
- (3) Ethidium bromide
- (4) Bromophenol blue



44. Select the correct answer

- I. Restriction enzymes cut the strand of DNA a little away from the end of the palindrome sites, but between the same two bases on the opposite strands.**
- II. The overhanging stretches are called sticky ends on each strand**

- | | |
|-----------------------|------------------------|
| (1) Only I is correct | (2) Only II is correct |
| (3) Both are correct | (4) None are correct |



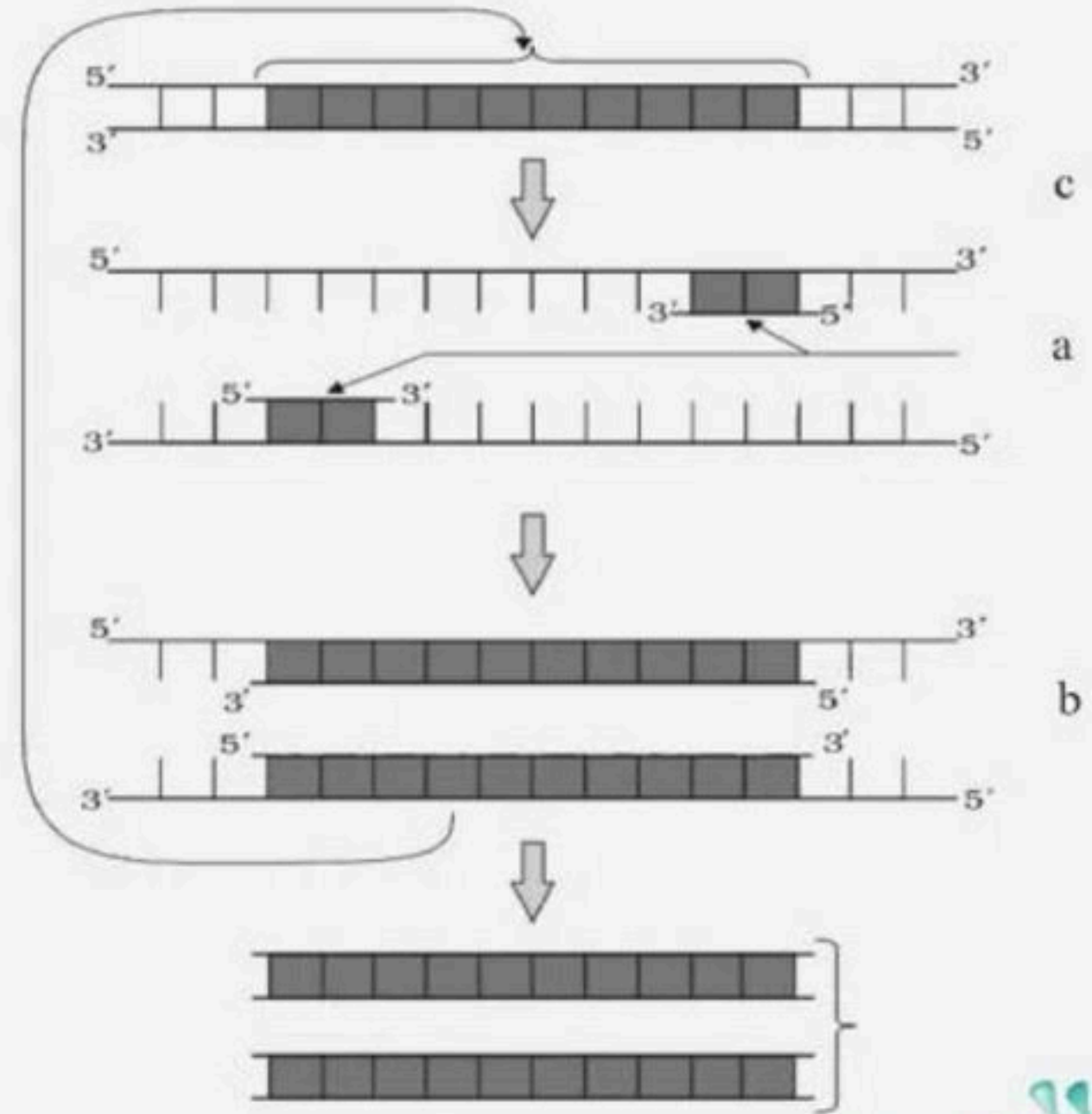
45. In gel electrophoresis :-

- (1) DNA fragments are arranged according to charge
- (2) DNA fragments are treated by alkali solution
- (3) DNA fragments are denatured
- (4) DNA fragments are arranged according to length on gel by applying electric field



46. Identify the processes labelled a,b,c in the following diagram.

- (1) a - annealing, b-denaturation, c- extension
- (2) a-extension b-denaturation, c - annealing
- (3) a - denaturation, b - extension, c - annealing
- (4) a - annealing, b-extension, c- denaturation



47. Upon exposure to UV radiation, DNA stained with ethidium bromide will show

- (1) Bright red colour
- (2) Bright blue colour
- (3) Bright yellow colour
- (4) Bright orange colour



48. Statement A : *Agrobacterium tumefaciens* is popular in genetic engineering because this bacterium is associated with roots of cereal and pulse crops.

Statement B : Retroviruses in animals including humans are able to change normal cells into cancerous cells

- (1) Statement A is correct but Statement B is incorrect.
- (2) Statement A is incorrect but Statement B is correct.
- (3) Both statements are correct.
- (4) Both Statements are incorrect.



49. The different basic steps of genetically modifying an organism are given below randomly

I. Identification of DNA with desirable genes

II. Maintenance of introduced DNA in the host and transfer of the DNA to its progeny.

III. Introduction of identified DNA into the host

Which of the following represents the correct sequence of steps?

(1) I, II, III

(2) I, III and II

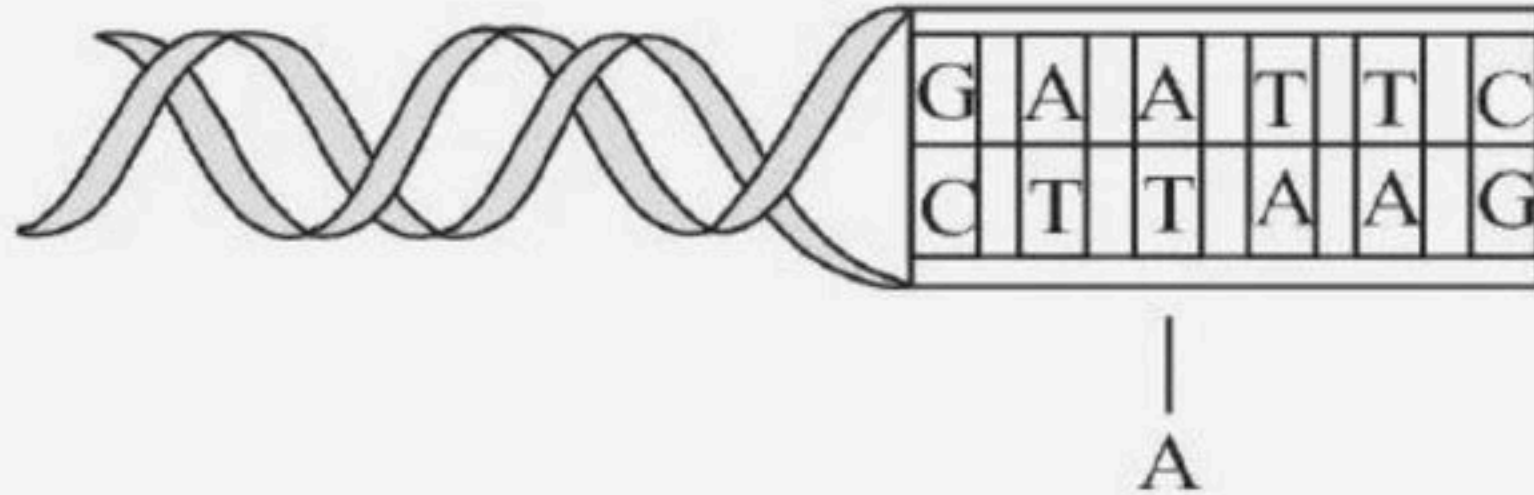
(3) III, II and I

(4) None of them



50. Identify the labelled part A

- (1) Restriction enzyme
- (2) Palindrome
- (3) Both 1 and 2
- (4) Cloning vector



51. Assertion : The matrix used in gel electrophoresis should have controllable pore size

Reason : Agarose concentration can be changed to change pore sizes

- (1) Both Assertion and Reason are correct and Reason is the correct explanation of the Assertion.
- (2) Both Assertion and Reason are correct but Reason is not the correct explanation of the Assertion.
- (3) Assertion is correct but Reason is incorrect.
- (4) Both Assertion and Reason are incorrect



52. Statement A : PCR technique is helpful in detecting bacterial and viral diseases even when symptoms of disease are not yet visible

Statement B : PCR is helpful in the identification of the coronavirus.

- (1) Statement A is correct but Statement B is incorrect.
- (2) Statement A is incorrect but Statement B is correct.
- (3) Both statements are correct.
- (4) Both Statements are incorrect.



53. The key tools required for recombinant DNA technology are

i. Restriction endonucleases

ii. Polymerase enzyme

iii. Ligase

iv. Vectors

v. Host organisms

(1) i, ii & iii

(2) i, iii, iv & v

(3) i, ii, iii & iv

(4) i, ii, iii, iv & v



54. Match the columns I and II.

Column I

A Palindromic nucleotide sequences

B Eco RI

C Hind II

(1) A – R , B – P , C – Q

(2) A – Q , B – R , C – P

(3) A – P , B – R , C – Q

(4) A – Q , B – P , C – R

Column II

P GAATTC

Q First endonuclease

R Site for cutting endonucleases



55. Agarose which finds use in gel electrophoresis is a ____

- (1) Lab made monomer
- (2) Natural polymer
- (3) Artificial polymer
- (4) Available in Earth's crust



Title: Biotechnology : Principle & Processes

56. Match the columns I and II

Column I

A DNA fragments separate according to their size

B Extracted of DNA from the gel

C Extraction of pure DNA

D Transformation

Column II

P Spooling

Q Introduction of DNA
in a host bacterium

R Elution

S Sieving effect

(1) A – R, B – P , C – Q, D – S

(2) A – S, B – R , C – P, D – Q

(3) A – P, B – R, C – Q, D – S

(4) A – Q, B – P , C – R , D – S



57. Which of the following statement is false about cloning vector.

- (1) 'Ori' is a sequence responsible for controlling the copy number of the linked DNA
- (2) Selectable marker selectively permitting the growth of the transformants
- (3) In order to link the DNA, the vector needs to have single recognition site for the commonly used restriction enzymes
- (4) Alien DNA is amplified by PCR



58. Assertion : In gel electrophoresis, DNA fragments are separated.

**Reason : DNA is negatively charged, so it moves
towards anode under electric field**

- (1) Both Assertion and Reason are correct and Reason is the correct explanation of the Assertion.
- (2) Both Assertion and Reason are correct but Reason is not the correct explanation of the Assertion.
- (3) Assertion is correct but Reason is incorrect.
- (4) Both Assertion and Reason are incorrect



59. The specific sequence of DNA that initiate replication of alien DNA for cloning of DNA is called as

- (1) Promoter
- (2) Origin of replication
- (3) Origin of combination
- (4) Initiator



60. Select the correct answer

- i. Exonucleases remove nucleotides from the ends of the DNA**
- ii. Endonucleases make cuts at specific positions within the DNA.**

- (1) Only i is correct
- (2) Only ii is correct
- (3) Both are correct
- (4) None are correct



61. The definition of biotechnology given by EFB encompasses

- (1) Traditional view
- (2) Modern molecular biotechnology
- (3) Hybridiation
- (4) Both (1) and (2)



62. Statement A : Origin of replication is an essential part of a vector and is responsible for initiating replication

Statement B : The vector requires a selectable marker, which helps in identifying and eliminating nontransformants and selectively permitting the growth of the transformants.

- (1) Statement A is correct but Statement B is incorrect. Sigma Institute
- (2) Statement A is incorrect but Statement B is correct.
- (3) Both statements are correct.
- (4) Both Statements are incorrect.



63. Statement A : *Agrobacterium tumefaciens* is a pathogen of several monocot plants.

Statement B : Retroviruses in plants have ability to transform normal cells into cancerous cell.

- (1) Statement A is correct but Statement B is incorrect.
- (2) Statement A is incorrect but Statement B is correct.
- (3) Both statements are correct.
- (4) Both Statements are incorrect.



64. Statement A : Recognition site should be preferably single and responsive to commonly used restriction enzyme.

Statement B : In pBR322 alien DNA is ligated generally in the area of Bam-HI site of ampicillin resistance gene.

- (1) Statement A is correct but Statement B is incorrect.
- (2) Statement A is incorrect but Statement B is correct.
- (3) Both statements are correct.
- (4) Both Statements are incorrect.



65. An antibiotic resistance gene in a vector usually helps in the selection of

- (1) Non-competent cells
- (2) Competent cells
- (3) Non-recombinant cells
- (4) Transformed cells



66. If we ligate a foreign DNA at the BamHI site in pBR322, the recombinant plasmid will :-

- (1) Show ampicillin resistance only.
- (2) Show tetracycline resistance.
- (3) Grow well on tetracycline containing medium.
- (4) Not grow on ampicillin containing medium



67. The components of a bioreactor are

- i. An agitator system.**
- ii. An oxygen delivery system**
- iii. Foam control system**
- iv. Temperature control system**
- v. control system**
- vi. Sampling ports**

Choose the correct option

- (1) i, ii, iii,
- (2) i, iii, iv and v
- (3) i, ii, iv, v, vi
- (4) All of these



68. Select the correct answer

I. In the year 1968, the two enzymes responsible for restricting the growth of bacteriophage in Escherichia coli were isolated.

II. One of these added ethyl groups to DNA, while the other cut DNA.

- (1) Only I is correct
- (2) Only II is correct
- (3) Both are correct
- (4) None are correct



69. Identify the process



(1) Elution

(2) Spooling

(3) Spooning

(4) Sieving



70. Normally, the genes encoding resistance to antibiotics such as ____ selectable markers for E. coli.

- (1) Ampicillin, chloramphenicol
- (2) Tetracycline, kanamycin
- (3) Amoxycillin, penicillin
- (4) Both 1 and 2



71. Read the statements

I. When cut by the same restriction enzyme, the resultant DNA fragments have the same kind of sticky ends and these can be join together using DNA ligases.

II. The bacterial/plant, animal cell is broken with RNA, proteins, polysaccharides and lipids

(1) I is true, but II is false

(2) I is false, but II is true

(3) I and II are true

(4) I and II are false



72. Statement A : Restriction enzymes recognize palindromic sequence.

Statement B : DNA is negatively charged, so it moves towards anode under electric field

- (1) Statement A is correct but Statement B is incorrect.
- (2) Statement A is incorrect but Statement B is correct.
- (3) Both statements are correct.
- (4) Both Statements are incorrect.



73. The normal E. coli cells do not carry resistance against any of these antibiotics. Which are these antibiotics?

- i. Ampicillin
- ii. Chloramphenicol,
- iii. Tetracycline
- iv. Kanamycin

- (1) i, ii
- (2) , ii, iii
- (3) i, ii, iii, iv
- (4) None of them



74. All the following statements about Stanley Cohen and Herbert Boyer are correct but one is wrong. Which one is wrong?

- (1) They first discovered the technique of gel electrophoresis .
- (2) They discovered recombinant DNA (rDNA) technology, and this marks the birth of modern biotechnology.
- (3) They isolated the antibiotic resistance gene by cutting out a piece of DNA from the plasmid which was responsible for conferring antibiotic resistance.
- (4) They invented genetic engineering by combining a piece of foreign DNA containing a gene from a bacterium with a bacterial plasmid using the enzyme restriction endonuclease.



75. Insertional inactivation is related to

i. Ampicillin

ii. Chloramphenicol,

iii. Tetracycline

iv. Kanamycin

(1) Permease gene

(2) Trans-acetylase gene

(3) Beta galactosidase

(4) Beta galactoside



76. Restriction endonuclease binds to DNA and cuts two strands of double helix at specific points in their

- (1) Glycoside bonds
- (2) Hydrogen bond
- (3) Sugar-phosphate backbone
- (4) N-glycosidic bonds



77. The first restriction endonuclease type II, P was isolated from Q bacterium.

P

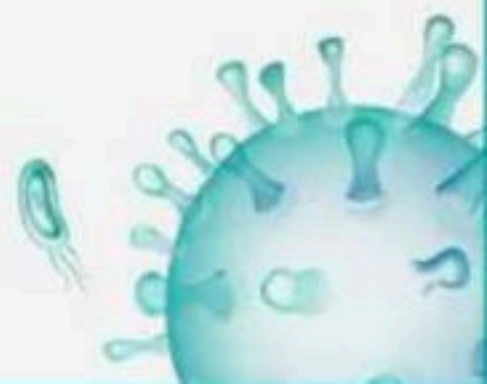
Q

- | | |
|--------------|----------------------------|
| (1) Eco RI | Escherichia RY13 |
| (2) Eco RIII | Escherichia coli |
| (3) Hind II | Haemophilus influenzae |
| (4) Bam HI | Bacillus amyloliquefaciens |



78. Primers are

- (1) Specific sequences that serve as restriction sites
- (2) Chemically synthesised oligonucleotides that are not complementary to the region of template DNA
- (3) The double-stranded DNA that need to be amplified
- (4) Small chemically synthesised oligonucleotides that are complementary to the region of template DNA



79. DNA polymerase can be obtained from

- (1) *Thermus aquaticus*
- (2) pBR 322
- (3) *Agrobacterium*
- (4) Retrovirus



80. Which is true about biotechnology techniques?

- (1) PCR is used to cut DNA molecules
- (2) A DNA probe consists of radioactive single strand of DNA
- (3) Restriction enzymes were first discovered in Bacteriophages
- (4) Pvu I is a name for a DNA probe



81. Statement A : Maintenance of sterile environment is essential for manufacture of biotechnological products.

Statement B : Agrobacterium tumefaciens delivers a piece of DNA known as 'A-DNA' which transforms normal plant cells into tumour cells and directs these tumour cells to produce chemicals against pathogens

- (1) Statement A is correct but Statement B is incorrect.
- (2) Statement A is incorrect but Statement B is correct.
- (3) Both statements are correct.
- (4) Both Statements are incorrect.



82. Which among the following is correct regarding a bioreactor

- i. It can process small volume of culture**
- ii. It provides optimum temperature, pH, salt, vitamins and oxygen**
- iii. sparged stirred tank bioreactor is a stirred type reactor which is air bubbled.**

- (1) i & ii
- (2) i & iii
- (3) ii & iii
- (4) i, ii & iii



83. Main steps in the formation of Recombinant DNA are given below. Arrange these steps in a correct sequence.

- A. Insertion of recombinant DNA into the host cell**
- B. Cutting of DNA at specific location by restriction enzyme**
- C. Isolation of desired DNA fragment**
- D. Amplification of gene of interest using PCR**

Choose the correct answer from the options given below

- | | |
|----------------|----------------|
| (1) B, C, D, A | (2) C, A, B, D |
| (3) C, B, D, A | (4) B, D, A, C |



84. Proteins are removed by treatment with

- (1) Ribonuclease
- (2) Chitinase
- (3) Cellulase
- (4) Protease



85. Ti plasmid is useful in genetic engineering by

- (1) Bringing new genes into animal cells
- (2) Bringing tumor cells into plant cells
- (3) Bringing useful and required genes into plants
- (4) All of the above



86. Assertion : The construction of the first recombinant DNA emerged from the possibility of linking a gene encoding antibiotic resistance with a native plasmid of *Salmonella typhimurium*.

Reason : Plasmid is an autonomously replicating circular extrachromosomal DNA so they are used as vectors

- (1) Both Assertion and Reason are correct and Reason is the correct explanation of the Assertion.
- (2) Both Assertion and Reason are correct but Reason is not the correct explanation of the Assertion.
- (3) Assertion is correct but Reason is incorrect.
- (4) Both Assertion and Reason are incorrect



87. What is the fate of a piece of DNA carrying only gene of interest which is transferred into an alien organism?

A. The piece of DNA would be able to multiply itself independently in the progeny cells of the organism.

B. It may get integrated into the genome of the recipient.

C. It may multiply and be inherited along with the host DNA.

D. The alien piece of DNA is not an integral part of chromosome.

E. It shows ability to replicate.

Choose the correct answer from the options given below:

(1) A and B only

(2) D and E only

(3) B and C only

(4) A and E only



88. Specific DNA sequence GAATTC is cut by which of the following restriction endonuclease

- (1) Hind II
- (2) Hind III
- (3) Eco RII
- (4) Eco RI



89. Palindrome sequences

- (1) Read same on two strands when orientation of reading is different
- (2) Read different on two strands when orientation of reading is same
- (3) Read same on two strands when orientation of reading is same
- (4) Read opposite on two strands when orientation of reading is same



90. Match the columns I and II.

Column I		Column II	
A	Insertional inactivation	P	RNA to DNA
B	<i>Agrobacterium tumefaciens</i>	Q	Colourless colonies
C	Retroviruses	R	T-DNA
		S	A-DNA
		T	Blue colonies

- (1) A – R, B – P, C – Q,
(2) A – Q, B – R, C – P,
(3) A – P, B – R, C – Q,
(4) A – Q, B – P, C – R



91. Match the columns I and II.

Column I		Column II	
A	Stanley Cohen and Herbert Boyer	P	1968
B	Two enzymes responsible for restricting the growth of bacteriophage	Q	RY 13
C	Hind II	R	1963
D	EcoRI	S	1972

- (1) A – R, B – P, C – Q, D – S
(2) A – S, B – R, C – P, D – Q
(3) A – P, B – R, C – Q, D – S
(4) A – Q, B – P, C – R, D – S



92. Match the columns I and II.

Column I		Column II	
A	Lysozyme	P	Fungus
B	Cellulase	Q	Protease
C	Chitinase	R	Plant cell
D	RNA	S	Bacteria
E	Proteins	T	Ribonuclease

- (1) A – R, B – P, C – Q, D – T, E – S
(2) A – Q, B – R, C – P, D – T, E – S
(3) A – S, B – R, C – P, D – T, E – Q
(4) A – Q, B – T, C – P, D – R, E – S



93. Match the columns I and II.

Column I		Column II	
A	PCR P	P	Desired protein
B	Bioreactors	Q	Separation and purification
C	Recombinant protein	R	Huge culturing of organisms
D	Downstreaming process	S	Amplification of DNA

- (1) A – R, B – P, C – Q, D – S
(2) A – S, B – R, C – P, D – Q
(3) A – P, B – R, C – Q, D – S
(4) A – Q, B – P, C – R, D – S



94. Match the following

Column I		Column II	
A	Endonuclease	P	Joins two ends of DNA
B	Plasmid	Q	Make cut at specific positions
C	DNA ligase	R	Remove nucleotides from ends of DNA
D	Exonuclease	S	Autonomously replication self replicating circular extra chromosomal DNA

- (1) A - Q, B - S, C - P, D - R
- (2) A - P, B - R, C - S, D - Q
- (3) A - Q, B - P, C - R, D - S
- (4) A - S, B - Q, C - P, D - R



95. Match the columns I and II.

Column I		Column II	
A	Insertional inactivation	P	Transform normal cell into cancerous cells
B	<i>Agrobacterium tumifaciens</i>	Q	Beta-galactosidase gene
C	Retroviruses	R	Pathogen of several dicot plants

- (1) A – R, B – P, C – Q,
- (2) A – Q, B – R, C – P,
- (3) A – P, B – R, C – Q,
- (4) A – Q, B – P, C – R



96. Match the following

Column I		Column II	
A	<i>Thermus aquaticus</i>	P	Plasmid
B	pBR322	Q	DNA Polymerase
C	Ligase	R	Restriction endonuclease
D	<i>Hind II</i>	S	Molecular glue

- (1) A - Q, B - P, C - S, D - R
(2) A - P, B - S, C - R, D - Q
(3) A - R, B - Q, C - S, D - P
(4) A - Q, B - S, C - P, D - R



97. Match the columns I and II.

Column I		Column II	
A	Exonucleases	P	Make cuts at specific positions in the DNA
B	Endonucleases	Q	Joins two DNA
C	Ligase	R	Remove nucleotides from the ends of the DNA

- (1) A – R, B – P, C – Q,
- (2) A – Q, B – R, C – P,
- (3) A – P, B – R, C – Q,
- (4) A – Q, B – P, C – R



98. Match the columns I and II.

Column I		Column II	
A	Recombinant DNA	P	Use of gene cloning and gene transfer
B	Origin of replication	Q	Circular DNA
C	Plasmid	R	Initiating replication

- (1) A – R , B – P , C – Q
- (2) A – Q , B – R , C – P
- (3) A – P , B – R , C – Q
- (4) A – Q , B – P , C – R



99. Match the organism with its use in biotechnology
Select the correct option from the following:

Column I		Column II	
A	Bacillus thuringiensis	P	Cloning vector
B	Thermus aquaticus	Q	Construction of first DNA molecules
C	Agrobacterium tumefaciens	R	DNA polymerase
D	Salmonella typhimurium	S	Cry proteins

- (1) A – S, B – R, C – P, D – Q
- (2) A – R, B – Q, C – S, D – P
- (3) A – R, B – S, C – P, D – Q
- (4) A – Q, B – S, C – R, D – P



100. Match the columns I and II wrt E. coli cloning vector pBR322.

Column I		Column II	
A	BamH I	P	ampR
B	Pst I	Q	tetR
C	Ori	R	Origin of replication
D	Rop	S	Proteins involved in the replication of the plasmid.

- (1) A – R, B – P, C – Q, D – S
- (2) A – S, B – R, C – P, D – Q
- (3) A – P, B – R, C – Q, D – S
- (4) A – Q, B – P, C – R, D – S



THANK YOU



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